

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	18 July 2026
Team ID	
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	EV Data Collection	Import EV vehicle data from datasets Collect charging station information
FR-2	EV Data Processing	Clean and preprocess EV data Handle missing and inconsistent data
FR-3	View EV Details	Display EV specifications such as brand, model, range, efficiency, and price Provide detailed information about selected EV models
FR-4	EV Comparison	Compare multiple EV models based on range, price, and features Provide comparison charts and visual insights
FR-5	Charging Station Analysis	Display charging station locations and details Analyze charging infrastructure data
FR-6	Data Visualization Dashboard	Generate interactive charts and dashboards Provide filters for better data exploration

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard should be simple, user-friendly, and easy to understand.
NFR-2	Security	User data and system information should be protected from unauthorized access.
NFR-3	Reliability	The system should provide accurate and consistent EV analysis results.
NFR-4	Performance	The dashboard should load visualizations and data efficiently.
NFR-5	Availability	The system should be accessible whenever users need EV insights.
NFR-6	Scalability	The system should support additional EV datasets and future enhancements.